JK Lakshmipat University Jaipur
INSTITUTE OF ENGINEERING AND TECHNOLOGY
Mid Term Examination, October 2026
B. Tech., Semester I (Sections D and H), 2026-27

Roll No. 579

CS1139: Programming - I

Time: 1.5 hrs Max. Marks: 20

Instructions to students:

1. Do not write anything other than your roll number on question paper.
2. Mention all the assumptions for your answers clearly.
3. **Calculators are not allowed.**

Q. 1.1 What will be the output of the following code? 1.0 *(CO3, BTL3)*

```
x = 8
if x > 5:
    x = x * 2
    if x > 10:
        x = x - 3
        if x % 2 == 0:
            print("Even")
        else:
            print("Odd")
    else:
        print("Small")
else:
    print("Very Small")
```

Q. 1.2 What will be the output of the following code? 1.0 *(CO2, BTL2)*

```
a = True + True
b = False + 0.1
c = 0.2
print(a)
print(b + 0.2 == 0.3)
print((True * 5 + 0.1) == 5.1)
```

Q. 1.3 What will be the output of the following code? 1.0 *(CO2, BTL3)*

```
a,b,c = 2,4,3
print(a < b < c + 2 and not b % a == 1 or c ** a >= b + 4)
```

Q. 1.4 What will be the output of the following code? Justify your answer. 1.0 *(CO2, CO3, BTL4)*

```
i = 1
while True:
    if i > 4:
        break
    if i == 2:
        i += 1
        continue
    print(i)
    i += 1
```

Q. 2.1 The following code finds all factors of the number entered by the user, counts the factors, and determines if it is prime. Replace each ??? with the correct code. 2.0 *(CO1, CO3, BTL3)*

```
n = int(input("Enter a number: "))
count = 0
print("Factors:")
for i in range(1, n + 1):
    if ???:
        print(i)
        count += 1
print("Number of factors =", count)
if ???:
    print(n, "is prime")
else:
    print(n, "is not prime")
```

Q. 2.2 The following code checks whether the 3-digit number entered by the user is an Armstrong number or not. (A 3-digit Armstrong number is a number whose digits, each raised to the power 3, add up to the original number.) Replace each ??? with the correct code. 2.0 *(CO1, CO3, BTL3)*

```
n = int(input("Enter a number: "))
original = n
sum = 0
while n > 0:
    ???
    sum = sum + digit ** 3
    ???
if sum == original:
    print(original, "is an Armstrong number")
else:
    print(original, "is not an Armstrong number")
```

Q. 3 Create a **flowchart** and write the **pseudocode** for a process that takes a positive integer N as input and examines all the integers from 1 to N. For each number: 4.0 *(CO1, BTL6)*

- If the number is even, check whether it is also divisible by 4.
- If the number is odd, check whether it is divisible by 3.

Display the appropriate category for each number:

- Even and Divisible by 4
- Even but Not Divisible by 4
- Odd and Divisible by 3
- Odd but Not Divisible by 3

Finally, display the total count of numbers in each category.

Q.4.1 What will be displayed when the following Python program is executed? 2.0 *(CO2, CO3, BTL4)*

```
a = 5
b = 7
c = 3
if a + b * c > 15 and not b % c == 1 or a // c == 2:
    if a ** 2 - b * c >= 35:
        print("A")
    elif a + b % c * 2 > 8:
        print("B")
    else:
        print("C")
else:
    if (a * c - b ** 2 < 0 and not a % b == 3):
        print("D")
    elif a + c // b > b:
        print("E")
    else:
        print("F")
```

Q.4.2 Write a Python program to determine the bonus category of an employee based on years of service and performance rating. Input the employee's years of service and performance rating. Apply the following conditions: 2.0 *(CO1, CO3, BTL3)*

- If years of service is less than 0 or performance rating is outside the range 1 to 5, print "Invalid Input."
- If years of service is 10 or more and performance rating is 4 or above, print "Eligible for 20% Bonus."
- If years of service is 5 or more and performance rating is 3 or above, print Eligible for "10% Bonus."
- If performance rating is 5, print "Eligible for 15% Bonus".
- Otherwise, print "No Bonus".

Q. 5.1 In a Super Over, a team needs 15 runs from 6 balls to win. The scorer's program works as follows: 2.0 (CO1, CO3, BTL6)

- The runs scored on each ball are entered one at a time.
- Valid runs on a ball are 0, 1, 2, 3, 4 or 6. Any other value prints "Invalid entry" and is not counted as a ball.
- After each valid ball, print the runs still needed and the balls remaining.
- If the target is reached, print "Won with X balls to spare" and stop.
- If all 6 balls are bowled without reaching the target, print "Fell short by Y runs".

Write the program.

Q.5.2 Before a launch at Sriharikota, ISRO conducts a countdown rehearsal. Whenever a sensor check fails, the countdown is paused ("Hold") and a fresh attempt begins. The rehearsal log is generated by the following code: *2.0 (CO3, BTL3)*

```
for attempt in range(1, 4):
    for t in range(6, 0, -1):
        if t == 3 * attempt:
            print("Hold", end=" ")
            break
        print(t, end=" ")
    else:
        print("Liftoff!", end=" ")
    print()
```

(i) Write the exact output.

(ii) How many countdown numbers are printed in total, and in which attempt does liftoff happen?